

3.8 (a) Let's assume $X > 0$, $a.e.(P)$, then $1 = \{E(X^{1/2} \cdot 1/X^{1/2})\}^{1/2} \leq EXE(1/X)$, by **Cauchy-Schwarz** inequality, with equality holds iff $X = c$, $a.e.(P)$. This says X and $1/X$ are always negatively correlated unless they are constants.

(b) $f(x)g(x) \geq 1 \Rightarrow f(X) \geq 1/g(X)$, $a.e.(P) \Rightarrow E(f(X))E(g(X)) \geq E(1/g(X))E(g(X)) \geq 1$.

3.9 (i) Let $\Omega = \mathbb{R}$, $\mathcal{F} = \mathcal{B}(\mathbb{R})$ and μ be a probability measure with $\mu(\{i\}) = \frac{c_i}{\sum_{i=1}^k c_i}$, $i = 1, \dots, k$. Let $f(i) = a_i$, $g(i) = b_i$, $i = 1, \dots, k$ and $f(x) = g(x) = 0$, $x \notin \{1, \dots, k\}$. It's clear that $f \in L^p(\Omega, \mathcal{F}, \mu)$ and $g \in L^q(\Omega, \mathcal{F}, \mu)$. By Theorem 3.1.11, we obtain

$$\sum_{i=1}^k |a_i b_i| \frac{c_i}{\sum_{i=1}^k c_i} \leq \left(\sum_{i=1}^k |a_i|^p \frac{c_i}{\sum_{i=1}^k c_i} \right)^{\frac{1}{p}} \left(\sum_{i=1}^k |b_i|^q \frac{c_i}{\sum_{i=1}^k c_i} \right)^{\frac{1}{q}}.$$

That is,

$$\sum_{i=1}^k |a_i b_i| c_i \leq \left(\sum_{i=1}^k |a_i|^p c_i \right)^{\frac{1}{p}} \left(\sum_{i=1}^k |b_i|^q c_i \right)^{\frac{1}{q}}.$$

(ii) Let $p = q = 2$, then inequality (1.17) holds as a special case of (1.16).

3.17 (a) Let $p = p_2/p_1$ ($p > 1$), $1/p + 1/q = 1$, $Y = |X|^{p_1}$, then by Hölder's inequality,

$$E|X|^{p_1} = E(Y \cdot 1) \leq (EY^p)^{1/p} \{E(1^q)\}^{1/q} = \{E(Y^p)\}^{1/p} = (E|X|^{p_2})^{p_1/p_2}.$$

(b) The equality of the inequality in (a) holds iff $Y = c$, $a.e.(\mu)$, i.e. $|X|$ is a constant $a.e.(\mu)$.

3.18 (a) We'll assume that everything in the inequality is well-defined, because $\phi(x) = x \log x$ is convex in $(0, \infty)$, the result comes immediately from Jensen's inequality $E\phi(X) \geq \phi(EX)$.

However, if we think about the weakest condition on X to make this inequality true. $EX < \infty$ is a necessary and sufficient condition here for nonnegative X . It's necessary because if $EX = \infty$, $E \log X$ would not even be defined. The sufficiency is due to the Jensen's Inequality from *Real Analysis and Probability* (Richard M. Dudley (2004)).

(b) Let $\phi(x) = \sqrt{1+x^2}$, then $\phi(x)$ is convex, so $\sqrt{1+(EX)^2} = \phi(EX) \leq E\phi(X) = E\sqrt{1+X^2}$ given that $EX < \infty$. When $EX = \infty$, this is automatically true. As $1+X^2 \leq (1+X)^2$, so $E\sqrt{1+X^2} \leq E(1+X) = 1+EX$.